What happens at the horizon of an extreme black hole?

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Lucietti and HSR arXiv:1208.1437
Lucietti, Murata, HSR and Tanahashi arXiv:1212.2557
Murata, HSR and Tanahashi, to appear

Introduction

- Extreme black hole: zero Hawking temperature (surface gravity)
- e.g. $M = |Q|$ Reissner-Nordstrom, $M = \sqrt{|J|}$ Kerr
- Supersymmetric black holes necessarily extreme
- Are extreme black holes classically stable? Does a small initial perturbation remain small?

Supersymmetry vs stability

- Supergravity BPS bound: $M \geq |Q|$, supersymmetric (BPS) solutions saturate this
- Minimum energy $\Rightarrow$ stability?
- Maybe for field theory in flat spacetime
- Not with dynamical gravity e.g. nonlinear instability of AdS
  Bizon & Rostworowski 2011
- Not even for linear perturbations of a fixed black hole spacetime

Stability of black holes

- Consider Kerr solution
- Initial surface Σ extending from future event horizon $\mathcal{H}^+$ to infinity
- Kerr solution arises from initial data on Σ
- Perturb this data: expect small enough perturbation to disperse and spacetime will "settle down" to new Kerr solution (with perturbed parameters)
- No proof, even for linearized perturbations
- Best result: no exponentially growing "modes" $\sim e^{-i\omega t}R(r)\Theta(\theta)e^{im\phi}$ (Whiting 1989)

Black hole stability Dafermos & Rodnianski 2008-2010

- Schwarzschild or *non-extreme* Kerr black hole
- Toy model for linearized gravitational perturbations: massless scalar field

$$\Box\psi = 0$$

- Prescribe initial data for ψ on spacelike surface intersecting future event horizon $\mathcal{H}^+$ ($\psi \to 0$ at infinity)
- ψ and all its derivatives *decay* outside $\mathcal{H}^+$ and in a neighbourhood of $\mathcal{H}^+$

Killing Energy: Schwarzschild

- Timelike Killing field k^a gives conserved energy-momentum current $J^a = -T^a{}_b k^b$
- Killing energy of ψ on Σ: $E = \int_\Sigma J^a d\Sigma_a$, (non-negative, non-increasing in time)
- Try to use E to control ψ
- Problem: outgoing photons in $\mathcal{H}^+$ have zero Killing energy $\leftrightarrow$ energy density degenerates at $\mathcal{H}^+$ (doesn't control derivative of ψ transverse to $\mathcal{H}^+$)

Horizon redshift effect

- Horizon redshift effect: energy of photons in $\mathcal{H}^+$ measured by infalling observer redshifts as $e^{-\kappa v}$ ($\kappa =$ surface gravity, $v =$ Killing time along $\mathcal{H}^+$)
- Wave analogue used to prove decay of problematic derivative of ψ near $\mathcal{H}^+$
- Extreme black hole: $\kappa = 0$ so horizon redshift effect is absent
- Energy of outgoing photons at $\mathcal{H}^+$ does not decay
- Can't prove decay of transverse derivative of ψ at $\mathcal{H}^+$

Extreme RN: stability Aretakis 2011

- Massless scalar field $\Box\psi = 0$ in extreme Reissner-Nordstrom
- Stability result: ψ decays on, and outside $\mathcal{H}^+$

Extreme RN: conserved quantity Aretakis 2011

- Extreme RN: use ingoing Eddington-Finkelstein coordinates: regular at $\mathcal{H}^+$
- Assume spherical symmetry, wave eq. $\Box\psi = 0$ becomes $(M = 1)$

$$2\partial_v\partial_r(r\psi) + \partial_r\left((r-1)^2\partial_r\psi\right) = 0$$

- Evaluate at $r = 1$: $\partial_v\partial_r(r\psi)|_{r=1} = 0$
- So we have a conserved quantity on $\mathcal{H}^+$:

$$H_0[\psi] \equiv \partial_r(r\psi)|_{r=1}$$

Extreme RN: non-decay Aretakis 2011

- $H_0[\psi] = (\partial_r \psi + \psi)_{r=1}$ conserved
- $\psi \to 0$ as $v \to \infty \Rightarrow \partial_r \psi$ generically does not decay at $\mathcal{H}^+$
- $T_{rr} = (\partial_r \psi)^2 \Rightarrow$ energy-momentum tensor at $\mathcal{H}^+$ does not decay
- Summary: absence of redshift effect $\Rightarrow$ *outgoing waves at* $\mathcal{H}^+$ *do not decay*

Extreme RN: instability Aretakis 2011

- r-derivative of wave eq. $\Rightarrow$

$$\partial_v \left[\partial_r^2 (r\psi)\right]_{r=1} = -(\partial_r \psi)_{r=1} \to -H_0$$

- Hence $\left[\partial_r^2 (r\psi)\right]_{r=1} \sim -H_0 v$ as $v \to \infty$
- Similarly $\partial_r^k \psi \sim H_0 v^{k-1}$
- Second and higher transverse derivatives of ψ at $\mathcal{H}^+$ generically *blow-up* at late time: instability
- Interpretation: $\partial_r \psi$ decays *outside* $\mathcal{H}^+$ but not *on* $\mathcal{H}^+$ hence $\partial_r^2 \psi$ becomes large at late time on $\mathcal{H}^+$
- *Polynomial*, not exponential, time-dependence
- (Numerical results)

Higher partial waves Aretakis 2011

- ℓth partial wave ψ_ℓ: conserved quantity $H_\ell = \partial_r^\ell \left[r\partial_r \left(r\psi_\ell \right) \right]_{r=1}$
- $\Rightarrow \partial_r^{\ell+1}\psi_\ell$ generically does not decay at $\mathcal{H}^+$, $\partial_r^{\ell+2}\psi_\ell$ generically blows up at late time on $\mathcal{H}^+$
- s-wave instability is strongest (involves fewest derivatives)

Instability in a supersymmetric theory

- Extreme RN is BPS solution of minimal $N = 2$ supergravity but this has no scalar field
- Type II supergravity compactified on T^6 has 4-charge BPS black hole solutions
- These reduce to extreme RN for equal charges
- Moduli fields constant in background: fluctuations are massless scalars
- Aretakis instability can be embedded in supersymmetric theory

Extreme Kerr instability Aretakis 2011-2012

- Restrict to *axisymmetric* massless scalar ψ - no superradiance
- Stability result: ψ decays on, and outside $\mathcal{H}^+$
- Extreme Kerr not spherically symmetric yet evaluating $\Box\psi = 0$ at $\mathcal{H}^+$ and projecting onto spherical harmonics gives infinite set of conserved quantities analogous to $H_\ell[\psi]$
- Transverse derivative of ψ at $\mathcal{H}^+$ generically does not decay
- Second and higher transverse derivatives of ψ at $\mathcal{H}^+$ generically blow up at late time: instability

General extreme black hole Lucietti & HSR 2012

- $\Box\psi = 0$ in arbitrary extreme black hole ($\mathcal{H}^+$ has compact cross-sections)
- Use "improved" Gaussian null coordinates near horizon
- $\exists$ Conserved quantity analogous to Aretakis' H_0
- Generic non-decay of transverse derivative of ψ at $\mathcal{H}^+$
- Blow-up of second transverse derivative assuming black hole has an AdS_2 in near-horizon geometry (true for all known extreme black holes)

AdS_2 calculation

- Extreme RN has $AdS_2 \times S^2$ near-horizon geometry:

$$ds^2 = -r^2 dv^2 + 2dvdr + d\Omega^2$$

- Aretakis argument applies here too - instability?
- But massless scalar in AdS_2 is stable!
- Here the "instability" is a coordinate effect

Massive scalar field Lucietti, Murata, HSR & Tanahashi 2012

- $\Box\psi - m^2\psi = 0$ in extreme RN, spherical symmetry
- If $m^2 = n(n+1)$ then can defined conserved quantities analogous to H_ℓ with $\ell = n \Rightarrow$ non-decay of $\partial_r^{n+1}\psi$ at $\mathcal{H}^+$ etc
- Instability for other values of m confirmed numerically
- Massive scalar is more stable

Extreme RN: gravitational and electromagnetic perturbations Lucietti, Murata, HSR & Tanahashi 2012

- Instability of massless scalar suggests possible instability of linearized gravitational/electromagnetic perturbations
- Gravitational and electromagnetic perturbations coupled
- Spherical harmonics $\ell = 1, 2, \ldots$ ("non-extreme" perturbation has $\ell = 0$: non-dynamical)
- Can decouple equations, construct conserved quantities
- $\ell = 2$: non-decay of gauge-invariant quantity at $\mathcal{H}^+$ involving 3 derivatives of metric/Maxwell potential perturbations
- Expect blow-up at late time on $\mathcal{H}^+$ of quantity with 4 derivatives

Extreme Kerr: linearized gravitational perturbations

Lucietti & HSR 2012

- Null tetrad $\{\ell, n, m, \bar{m}\}$
- Weyl tensor components: complex Newman-Penrose scalars Ψ_A, $A = 0, \ldots, 4$
- $\Psi_0 = C_{abcd}\ell^a m^b \ell^c m^d$, $\Psi_4 = C_{abcd} n^a \bar{m}^b n^c \bar{m}^d$,
- Perturb Kerr: $\delta\Psi_0$ and $\delta\Psi_4$ invariant under infinitesimal coordinate transformations and infinitesimal basis transformations
- Each satisfies *Teukolsky equation* with spin $s = 2, -2$
- Variation of parameters within Kerr family has $\delta\Psi_0 = \delta\Psi_4 = 0$

Teukolsky equation

- Restrict to axisymmetric perturbations
- Evaluate (derivatives of) Teukolsky eq. at $\mathcal{H}^+$, project onto spin-weighted spherical harmonics ${}_sY_{j(m=0)}$, $j \geq |s|$ (even though Kerr not spherically symmetric!)
- Obtain infinite set of conserved quantities labelled by $(s, j) \Rightarrow$ non-decay at $\mathcal{H}^+$ of quantities involving sufficiently many derivatives of $\delta\Psi_0$, $\delta\Psi_4$
- $j = 2 = -s$: non-decay of derivative of $\delta\Psi_4$ on $\mathcal{H}^+$
- Expect blow-up of second derivative of $\delta\Psi_4$ at late time on $\mathcal{H}^+$
- $\delta\Psi_0$ exhibits much weaker instability

Backreaction

- Is Aretakis instability present in nonlinear theory?
- What is "endpoint" of instability?

Nonlinear evolution (work in progress)

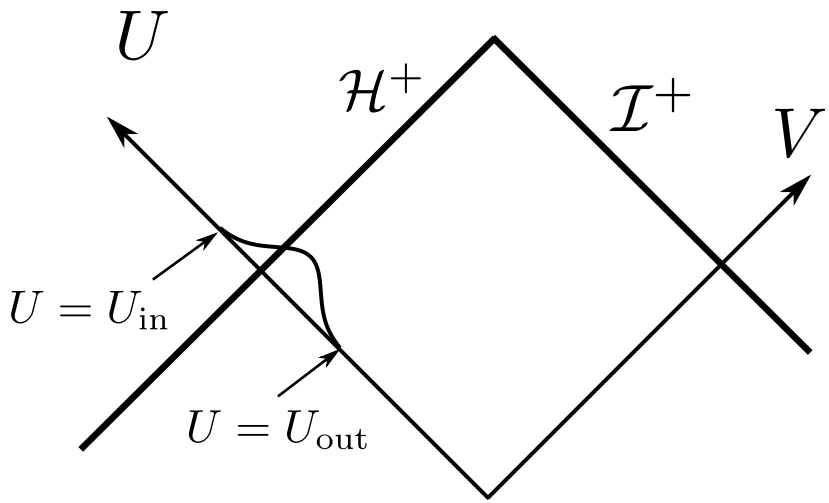
Murata, HSR & Tanahashi

- Model: Einstein-Maxwell theory coupled to massless scalar ψ assuming spherical symmetry
- Spherically symmetric metric in double null coordinates:

$$ds^2 = -f(U,V)dUdV + r(U,V)^2 d\Omega^2$$

- Maxwell field $\star F = Q d\Omega$ (Q is charge: conserved)
- Scalar field $\psi(U,V)$

Initial data

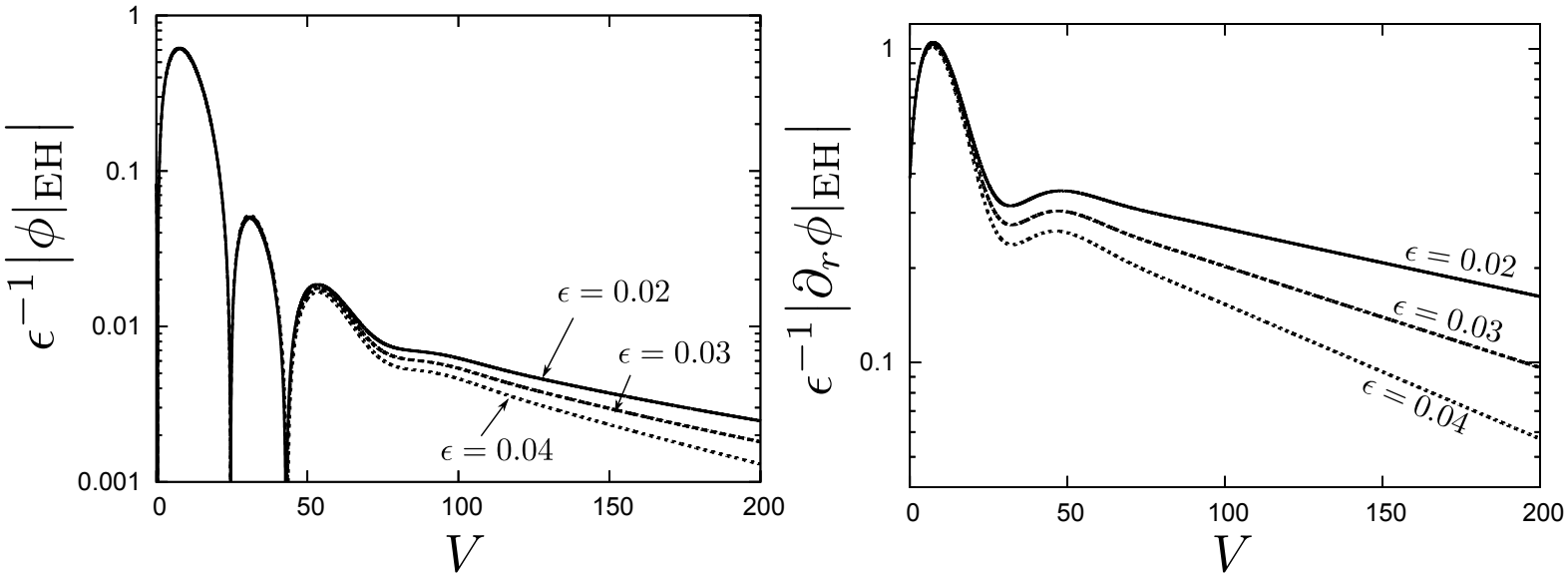


Initial data uniquely specified by outgoing wavepacket, amplitude ϵ, and initial Bondi mass M_i.
Data is RN except in $U_{\rm out} < U < U_{\rm in}$. Singularity at $r = 0$ OK if there is an event horizon $\Rightarrow M_i \geq Q$.

Initial data

- For given ϵ, how do we choose M_i?
- For large enough M_i, there are trapped surfaces behind an apparent horizon (trapped surface: ingoing *and* outgoing null geodesics normal to surface are converging)
- Reduce M_i so that data contains no trapped surfaces but still contains an apparent horizon: "degenerate apparent horizon", must have radius $r = Q$
- "Exterior" initial data is non-extreme RN

Results

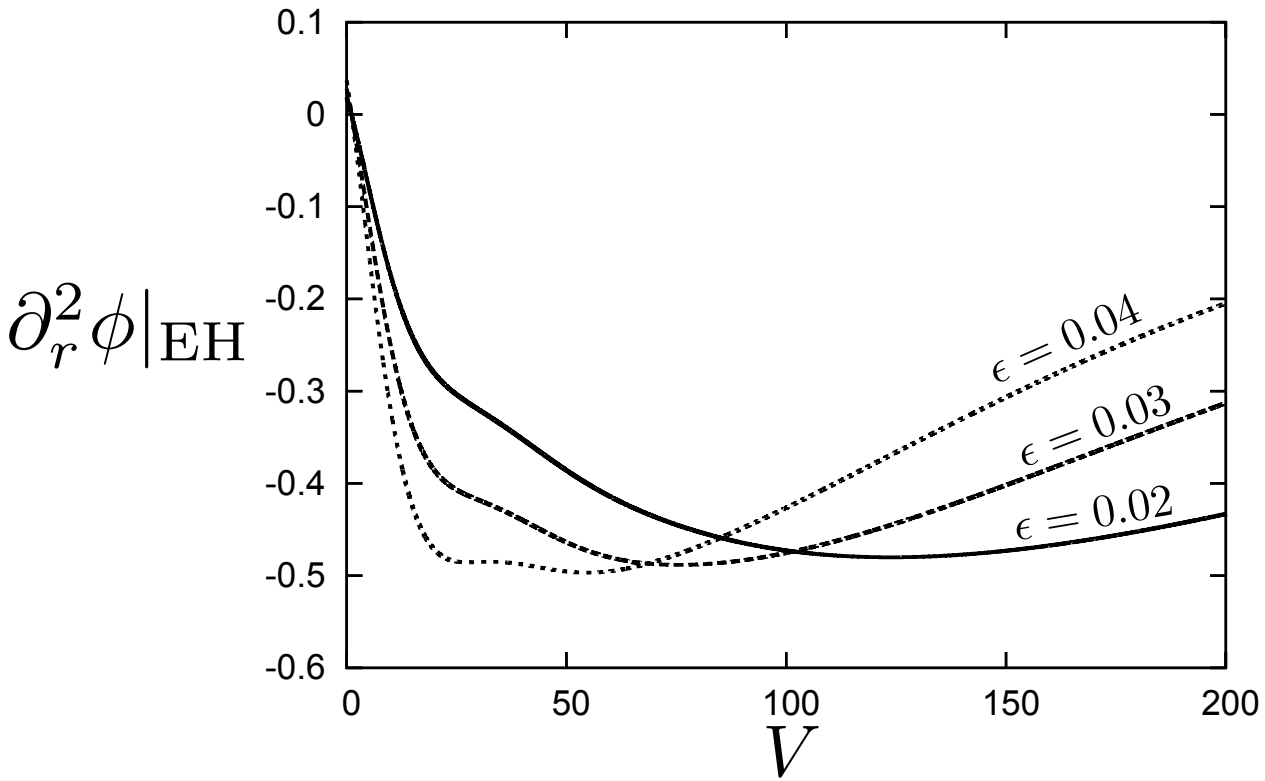

1
0.1
0.01
0.001
$\epsilon^{-1}|\phi|_{\mathrm{EH}}|$
$\epsilon = 0.02$
$\epsilon = 0.03$
$\epsilon = 0.04$
0
50
100
150
200
V
1
0.1
$\epsilon^{-1}|\partial_r\phi|_{\mathrm{EH}}|$
$\epsilon = 0.02$
$\epsilon = 0.03$
$\epsilon = 0.04$
0
50
100
150
200
V

Results

- Spacetime eventually settles down to a non-extreme RN black hole with $\kappa = \mathcal{O}(\epsilon)$
- For a time $V \sim 1/\epsilon$, the evolution is similar to the test field in extreme RN (gauge choice: $V \sim$ Eddington-Finkelstein)
- Slow decay $\sim e^{-\kappa V}$ of transverse derivative of field at horizon
- Linear growth of second transverse derivative until time $V \sim 1/\epsilon$, then slow decay

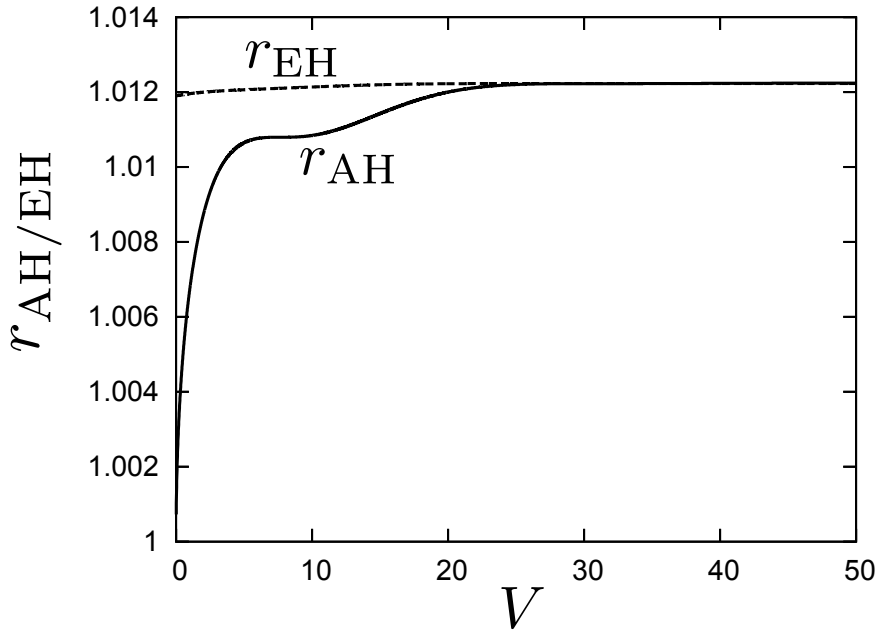
Nonlinear instability

Maximum value of second transverse derivative at horizon is $\mathcal{O}(1)$ as $\epsilon \longrightarrow 0$: instability!



Apparent and event horizons

Evolution of apparent horizon ($Q = 1$, $\epsilon = 0.05$)



Position of event horizon is $r = Q + \mathcal{O}(\epsilon)$

Toy model for back reaction

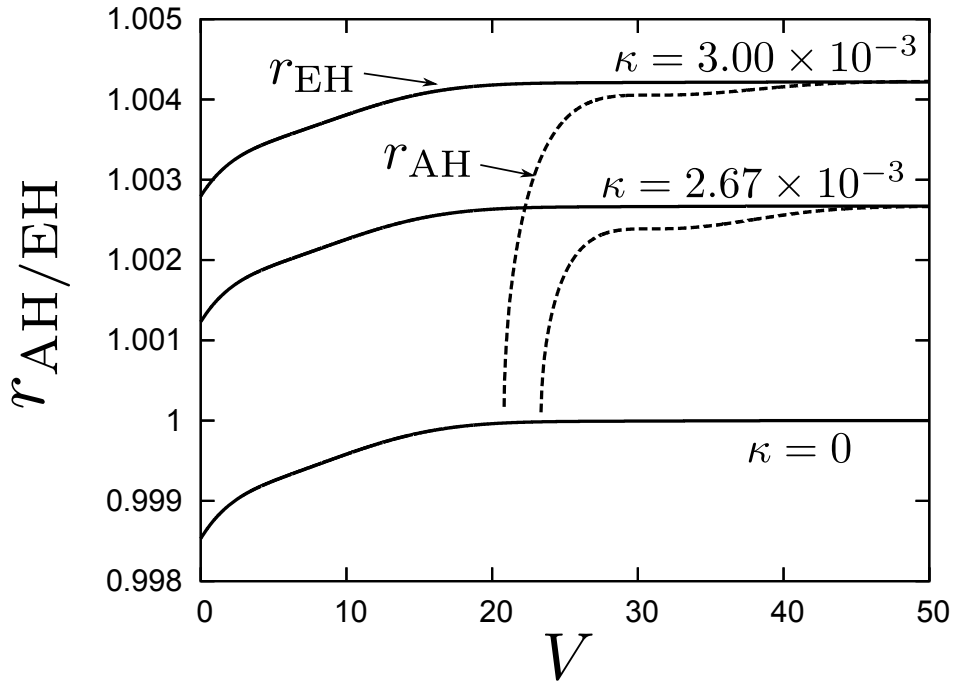
- Linear scalar field $\psi = \mathcal{O}(\epsilon)$ in *non-extreme* RN with $M = Q + \mathcal{O}(\epsilon^2)$
- Evaluate wave equation on $\mathcal{H}^+$: equation involving ψ and $\partial_r(r\psi)$. *Assume* $|\psi|$ bounded by its behaviour for extreme RN
- Find $\partial_r(r\psi)|_{\mathcal{H}^+}$ has slow exponential decay $\sim e^{-\kappa v}$ where surface gravity $\kappa = \mathcal{O}(\epsilon)$
- $\partial_r^2(r\psi)|_{\mathcal{H}^+}$ grows linearly to $\mathcal{O}(1)$ at time $v \sim 1/\epsilon$, slow exponential decay thereafter
- Agrees with numerical results

Dynamical extreme black holes

- Above initial data: no trapped surfaces but apparent horizon present. Trapped surfaces form in time evolution.
- Decrease M_i a little: no apparent horizon in initial data but trapped surfaces and apparent horizon form in time evolution
- Decrease M_i too much: no event horizon ("naked singularity")
- Critical value of M_i: event horizon but no trapped surfaces: *dynamical extreme BH* (definition)
- Third Law (Israel 1986): "non-extreme BH cannot become extreme"; this BH is always extreme

Dynamical formation of extreme black hole

Apparent horizon radius against time ($Q = 1, \epsilon = 0.1$)



Dynamical formation of extreme black holes

- Preliminary results indicate that solution approaches extreme RN outside $\mathcal{H}^+$ but scalar field on $\mathcal{H}^+$ behaves just as in linear theory: $\psi \to 0$, $\partial_r \psi \to H_0$, $\partial_r^2 \psi \sim H_0 v$
- "Final state" is extreme RN with $\psi = 0$ on and outside $\mathcal{H}^+$ but $\partial_r \psi = H_0$ on $\mathcal{H}^+$
- Energy-momentum tensor and curvature tensors discontinuous at $\mathcal{H}^+$
- H_0 is "hair" on the horizon?
- Entropy is same as for extreme RN

Summary

- Various test fields in extreme black hole spacetimes suffer an instability
- This instability persists in nonlinear theory, generically evolving to a non-extreme black hole
- Extreme black holes formed dynamically exhibit extra parameter(s) on horizon

Open questions

- CFT interpretation of conserved quantities
- Extreme RN/Kerr: *infinite* set of conserved quantities - for which extreme black holes do we have this?
- Interior structure of extreme black holes formed dynamically
- Formation of extreme black holes with charged scalar